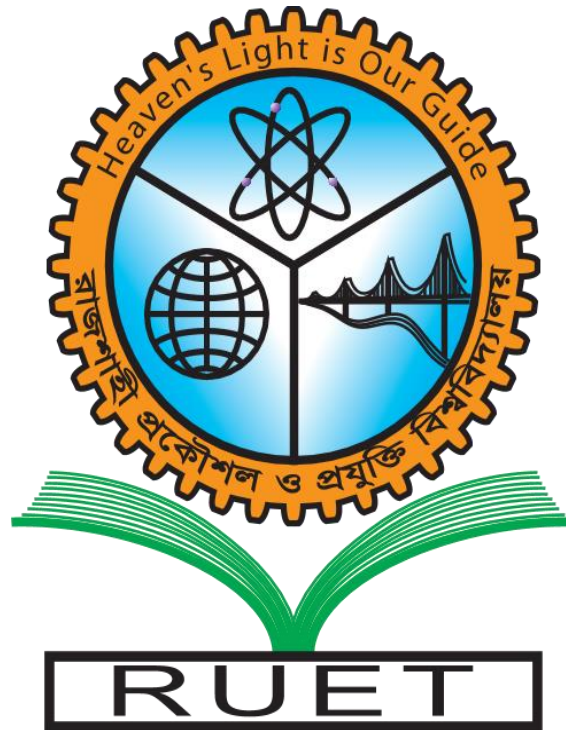


Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY



COURSE TITLE: Microprocessor, Assembly Language & Interfacing

COURSE CODE: ECE 3111

Assignment submission date: 23.11.2024

Submitted to:

Md. Omaer Faruq Goni
Assistant Professor,
Department of Electrical &
Computer Engineering, RUET.

Submitted by:

Puja Roy Dipa
Roll: 2010040
Session: 2020-2021

Introduction:

The advancements in embedded systems and microcontroller-based technology have made it possible to create cost-effective and efficient solutions for motor control. To control the speed of a DC motor using an 8086 microprocessor by generating a Pulse Width Modulation (PWM) signal. The PWM duty cycle is varied based on user input, allowing precise speed regulation.

The design incorporates several peripheral components, such as the 8255 Programmable Peripheral Interface (PPI), 8253 Programmable Interval Timer (PIT), ADC0808 (Analog-to-Digital Converter), and the L298 motor driver. A potentiometer is used as an input device to allow users to adjust motor speed in real time. The system converts the potentiometer's analog signal into a digital value using the ADC0808, dynamically modifying the PWM duty cycle to ensure precise and smooth motor operation.

This project highlights the capability of the 8086 microprocessor to interface seamlessly with peripheral devices, providing a practical and efficient solution for real-world motor control applications.

Objective:

The primary objectives of this project are as follows:

1. **Planning and Execution:**

To create a system that uses an 8086 microprocessor to generate a PWM signal that regulates a DC motor's speed.

2. **Peripheral Integration:**

To ensure smooth operation, integrate peripheral devices such as the 8255 PPI, 8253 PIT, ADC0808, and L298 motor driver.

3. **Speed Control in Real Time:**

This method uses an ADC to convert analog inputs to digital values and a potentiometer as user input to accomplish real-time motor speed variation.

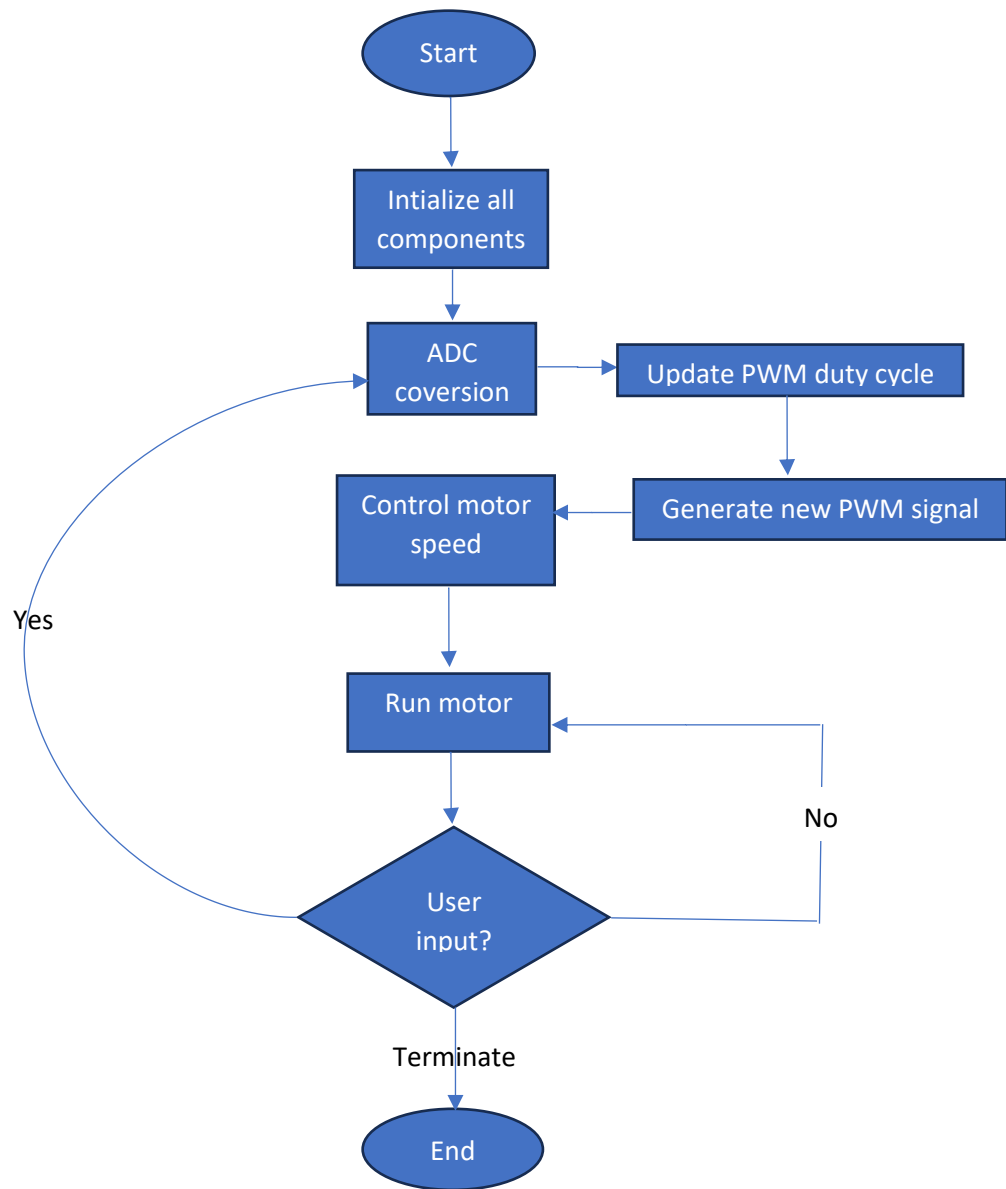
4. **Generation of Dynamic PWM:**

This ensures accurate and energy-efficient motor speed control by dynamically adjusting the PWM duty cycle in response to the digital signal from the ADC.

5. **System Illustration:**

To create a workable and effective solution that illustrates the possibilities of systems based on 8086 microprocessors for motor control applications.

Flow-chart:



Components using in this project:

8255 Programmable Peripheral Interface (PPI)

- **Purpose:** Acts as a bridge for communication between the microprocessor and peripheral devices by providing three configurable I/O ports: Port A, Port B, and Port C.
- **Usage in Code:**
 - Port A: Set as an output port to deliver control signals to the ADC and facilitate PWM signal generation.
 - Port B: Configured as an input port to receive digital data from the ADC post-conversion.

- Port C: Divided into two parts:
 - Upper Port (Input): Tracks ADC status signals, such as End-of-Conversion (EOC).
 - Lower Port (Output): Sends additional control signals as needed for the system.
- PPI_CNTRL: Used to configure port modes through control words.

8253 Programmable Interval Timer (PIT)

- **Purpose**: Enables precise timing and PWM signal generation through three independent counters.[1]
- **Usage in Code**:
 - Counter 0: Operates in Mode 2 (rate generator) to produce pulses for triggering system operations.
 - Counter 1: Configured in Mode 1 (Monostable Multivibrator) to generate single pulses with durations proportional to the ADC output.
 - PIT_CNTRL: Handles counter configuration, including mode selection and loading count values.

ADC0808 (Analog-to-Digital Converter)

- **Purpose**: Converts the analog voltage from the potentiometer into an 8-bit digital value to define the PWM duty cycle.[2]
- **Working Mechanism**:
 - Conversion is initiated by start and clock pulse.
 - The End-of-Conversion (EOC) signal is monitored using Port C.
 - Once the conversion is complete, the digital output is read through Port B.
- **Usage in Code**:
 - The ADC is reset and triggered using control signals from Port A.
 - The system waits for the EOC signal (detected through Port C) before accessing the converted digital value.

L298 Motor Driver

- **Purpose**: Controls the DC motor by amplifying the PWM signal and supplying the required power.
- **Usage in Code**:
 - The PWM signal generated by the 8255 PPI is sent to the L298 to manage motor speed.
 - The motor speed is directly regulated by the PWM duty cycle, which is determined by the value from Counter 1.

Potentiometer

- **Purpose**: Acts as an input device to generate an analog voltage that corresponds to the desired motor speed.

- **Working Mechanism:**

- The analog voltage from the potentiometer is fed into the ADC0808 for conversion into a digital value.
- This digital value is then used to dynamically adjust the PWM signal, altering the motor's speed.

Push Button

- **Purpose:** Provides user input to reset the ADC or initiate a new conversion cycle.
- **Usage in Code:**
 - Connected to the lower portion (PC7) of Port C on the PPI.
 - The button's state is monitored continuously within the **CHECK_BUTTON** routine.
 - When a press is detected, the ADC is reset, and a new conversion cycle begins.

Code:

DATA SEGMENT ;Defining port addresses FOR PPI_8255 AND PIT_8253

```
PPI_A EQU 80H
PPI_B EQU 82H
PPI_C EQU 84H
PPI_CNTRL EQU 86H

PIT_CNT0 EQU 60H
PIT_CNT1 EQU 62H
PIT_CNT2 EQU 64H
PIT_CNTRL EQU 66H
```

DATA ENDS

CODE SEGMENT

```
ASSUME CS:CODE, DS: DATA
MOV AX, DATA
MOV DS, AX
```

START:

```
MOV AL, 10001010B ; PORTA->Output, PORTB->Input, PORTC_Upper->Input, PORTC_Lower->Output
OUT PPI_CNTRL, AL
```

ATODCONVERTER:

```
MOV AL, 00H ;00000000
OUT PPI_A, AL
```

COUNTER:

```
MOV AL, 00110100B ;COUNTER 0, MODE2, Rate_Generator in hex, both msb anf lsb
OUT PIT_CNTRL, AL
```

```
MOV AL, 01h ;LSB 01
OUT PIT_CNT0, AL
```

```
MOV AL, 01h ;MSB 01; COUNT = 0101 = 101, in hex mode (FF+1)
OUT PIT_CNT0, AL
```

```
MOV AL, 01010010B ;Counter 1, MODE1, Monostable_Multivibrator, only LSB
OUT PIT_CNTRL, AL
```

CHANGE:

```
MOV AL, 00000000B ;Start pulse (just a spike) given in ADC
OUT PPI_C, AL
MOV AL, 00000001B
OUT PPI_C, AL
MOV AL, 00000000B
OUT PPI_C, AL
```

CHECK_EOC:

```
IN AL, PPI_C
AND AL, 00100000B
CMP AL, 00000000B
JE CHECK_EOC
```

LOOP_BACK:

```

IN AL, PPI_B
OUT PIT_CNT1, AL
CALL CHECK

```

```

CHECK PROC                ;BUTTON check function->If push button is pressed

```

```

IN AL, PPI_C
AND AL, 80H
CMP AL, 00H
JE CHANGE
CALL CHECK

```

```

CHECK ENDP

```

```

CODE ENDS

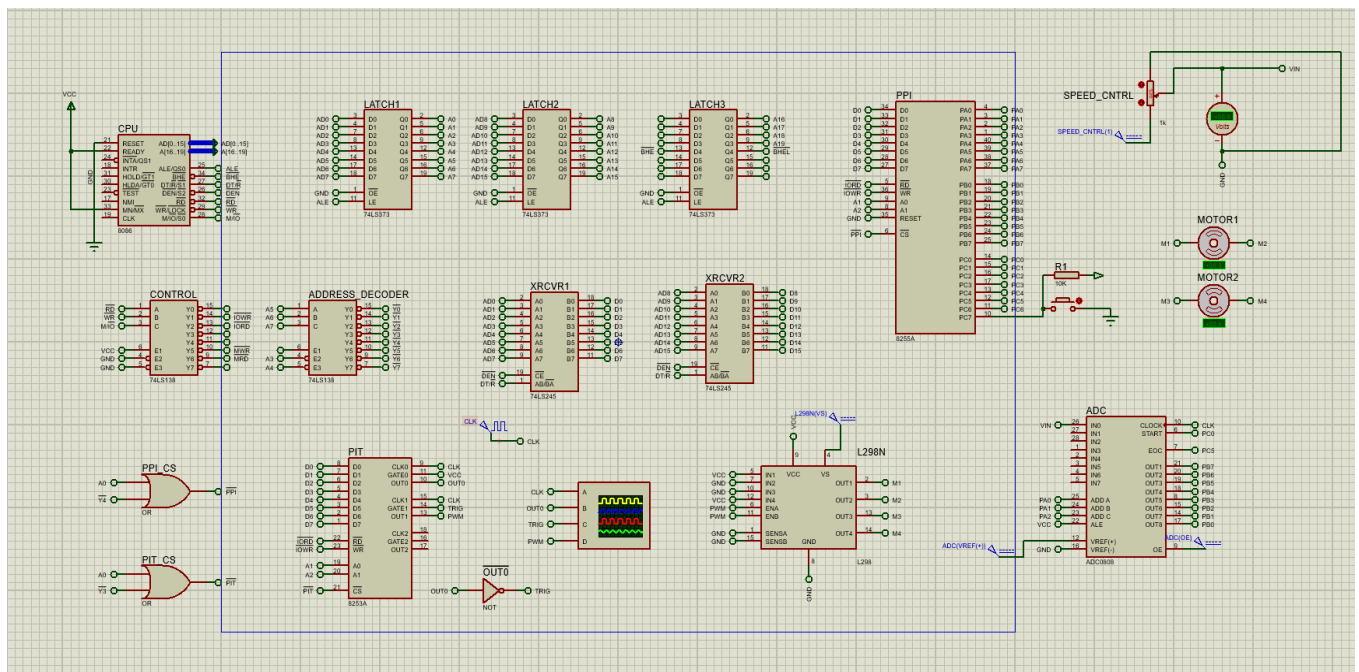
```

```

END START

```

Simulated circuit diagram:



Results:

The project demonstrates the successful design and development of an 8086 microprocessor-based system for PWM-based real-time DC motor speed control. The following outcomes are attained:

1. Precise Motor Speed Control

The PWM signal, which is dynamically generated based on the potentiometer's input, effectively regulates the speed of the DC motor. The motor's speed varies in real time based on the user-defined input (analog voltage) to the potentiometer.

2. Integration of Hardware Components

The system effectively integrates multiple peripheral devices, including the 8255 PPI, 8253 PIT, ADC0808, and L298 motor driver, ensuring seamless communication and coordination. Each component plays a crucial role in the overall functionality of the system:

1. The ADC0808 converts the analog input from the potentiometer into a corresponding digital value.
2. The 8253 PIT generates precise timing signals and PWM outputs.
3. The 8255 PPI manages data exchange between the microprocessor, ADC, and motor driver.
4. The L298 motor driver utilizes the PWM signal to control the DC motor's speed.

3. Real-Time User Input Management

The project enables real-time motor speed adjustments by continuously monitoring the potentiometer's voltage and dynamically modifying the PWM duty cycle. To enhance precision and responsiveness, a push button is incorporated, allowing the user to reset or restart the ADC conversion process when needed.

4. Efficient System Design

With minimal hardware requirements, the system offers an affordable solution for motor speed control, making it suitable for practical applications. Its modular architecture enhances scalability, allowing for seamless integration of additional sensors or expansion to control multiple motors.

5. Practical Applications

This project demonstrates an effective solution for motor speed regulation in applications such as robotics and industrial automation. It is equally applicable to consumer devices like fans, conveyor systems, and other appliances that require precise motor speed control.

References:

- [1] "DC motor speed control using PWM signals | Arrow.com." Accessed: Nov. 23, 2024. [Online]. Available: <https://www.arrow.com/en/research-and-events/articles/simple-dc-motor-speed-control-with-pwm>
- [2] "ADC0808 Pinout, How to Use & Datasheet." Accessed: Nov. 23, 2024. [Online]. Available: <https://components101.com/ics/adc0808-pinout-features-datasheet>

